

SOFTWARE REQUIREMENTS SPECIFICATION

Placement Eligibility and Resume Screening Portal

1. Introduction

1.1 Purpose

The purpose of this SRS is to define the functional and non-functional requirements of the Placement Eligibility and Resume Screening Portal. The system manages students, companies, placement drives, eligibility filtering, applications, resumes, deadlines, notifications, approvals, feedback, and placement reports. The system uses a centralized server, TCP socket communication, MySQL persistent storage, and role-based access.

1.2 Scope

The portal shall provide one centralized system through which students maintain placement information and resumes; recruiters register companies and provide job/eligibility criteria; TPO/TPC personnel manage placement activities and eligibility; Director and Dean users monitor placement information and participate in the defined communication/approval flow; and authorized users obtain placement reports and analytics.

1.3 Project Objectives

1. Centralize student, company, placement-drive and application information.
2. Maintain student academic details, skills and resumes for placement use.
3. Allow companies/recruiters to register and publish recruitment requirements.
4. Automatically filter students using predefined eligibility conditions.
5. Support application processing and placement-stage status tracking.
6. Manage placement-drive deadlines and provide system notifications.
7. Provide college reports, company reports and placement analytics.
8. Support concurrent clients through TCP socket-based client-server communication.

2. Overall Description

2.1 Product Perspective

The portal operates as a centralized multi-client system. Multiple authorized clients communicate with a central Java server through TCP sockets. The server performs authentication, authorization, validation and business processing, and accesses the MySQL database using JDBC. The logical flow is:

Client UI → TCP Socket → Central Server → Business Logic → JDBC → MySQL

2.2 Major System Modules

- Authentication and User/Role Management
- Student Profile and Resume Management
- Company Registration
- Placement Drive / Job Posting Management
- Eligibility Management
- Application Management
- Resume Screening
- TPO Approval and Drive Management
- TPC Eligibility Filtering
- Deadline and Notification Management
- Search and Candidate Filtering
- Feedback Management
- Reports and Analytics
- Audit Logging
- TCP Socket Communication
- Database Management

2.3 Operating Environment

The system is intended for a Java-based desktop/client-server environment. The implementation uses TCP sockets for communication and MySQL/JDBC for database persistence. Exact Java version, operating-system version and deployment topology shall be specified by the development team before final deployment.

2.4 Design and Implementation Constraints

- TCP Socket Programming is required for client-server communication.
- MySQL is the persistent relational database.
- Java/JDBC is used for application/database communication.
- The system must support multiple authenticated clients.
- Resume screening in the baseline system is rule/keyword based; advanced AI-based ranking is not part of the defined baseline.

3. Stakeholders and User Classes

User Class	Objective	Key Responsibilities / Access
Student	Participate in suitable placement opportunities.	Maintain profile/resume; view jobs; check eligibility; apply; track status; provide feedback.
Recruiter / Company HR	Conduct recruitment through the portal.	Register company; create job posting; define criteria/package/selection process; review candidates; shortlist.

TPO	Coordinate and control placement drives.	Review company/job details; Approve/Waiting/Reject; manage deadlines; monitor placement activity; reports.
TPC	Support student eligibility and coordination.	Filter eligible students; view eligible list/count; support drive activities.
Director	Review placement information and communication.	Read authorized placement information; message passing as permitted by final workflow.
Dean	Review placement information and communication.	Read authorized placement information; message passing as permitted by final workflow.
System Administrator	Maintain technical operation.	Manage users, permissions, server/database settings, logs and recovery.

3.1 Role Access Matrix

Function	Student	Recruiter	TPO	TPC	Director	Dean	Admin
Login	✓	✓	✓	✓	✓	✓	✓
Manage student profile	✓	—	—	—	—	—	✓
Upload/update resume	✓	—	—	—	—	—	✓
Register company	—	✓	—	—	—	—	✓
Create job/drive	—	✓	—	—	—	—	✓
Review/approve drive	—	—	✓	—	—	—	✓
Eligibility filtering	Own status	Own drives	✓	✓	✓*	✓*	✓
Apply to drive	✓	—	—	—	—	—	—
Resume screening	—	✓	As permitted	As permitted	—	—	✓
Reports/analytics	Limited	Own company	✓	As permitted	✓*	✓*	✓
Feedback	✓	✓	✓	✓	As permitted	As permitted	✓
User administration	—	—	—	—	—	—	✓

4. System Workflow

4.1 Student Workflow

1. Student logs in using valid credentials.
2. Student maintains PRN, course, CGPA, 10th percentage, 12th percentage and skills.
3. Student uploads or updates the resume.
4. Student views available job/placement postings and job descriptions.
5. System determines eligibility using the criteria configured for the placement drive.
6. Eligible student submits an application before the deadline.
7. System records the application and current status.
8. Student views application/selection status and relevant placement information.

4.2 Recruiter / Company Workflow

1. Recruiter registers the company and authenticates.
2. Recruiter creates a job/placement posting.
3. Recruiter enters job description, package, event details and deadline.
4. Recruiter defines eligibility criteria such as CGPA, course/department, passing year, backlogs and skills.
5. Recruiter defines selection stages such as GD, project and interview.
6. Posting is submitted for TPO review when the workflow requires approval.
7. After approval, the drive is made available to eligible students.
8. Recruiter views candidates, reviews resumes and performs shortlisting.

4.3 TPO/TPC Workflow

1. TPO receives a submitted company/job posting.
2. TPO verifies the company information, job details and eligibility criteria.
3. TPO sets the placement-drive status to Approved, Waiting or Rejected.
4. TPC applies eligibility filters and views the eligible-student list/count.
5. TPO monitors applications, deadlines and placement outcomes.

4.4 Director / Dean Workflow

1. Director/Dean logs in.
2. Director/Dean views authorized placement information.
3. Director/Dean reads placement-related details.
4. Director/Dean can pass messages to the concerned users.
5. The system records the message and its status.

4.5 End-to-End Workflow

Company Registration → Job/Drive Posting → Eligibility Criteria → TPO Review → Approved /Waiting / Rejected → Eligibility Filtering → Student Views Job → Application → Resume Screening / Shortlisting → Selection Status → Reports & Analytics

5. Functional Requirements

Requirement Name	Requirement Statement
User Authentication	The system shall authenticate a user before allowing access to protected functions.
Role Authorization	The system shall enforce permissions according to the authenticated user's role.
Student Profile	The system shall store and maintain PRN, course, CGPA, 10th, 12th, skills and other approved placement details.
Resume Management	The system shall allow a student to upload, replace and update a resume.
Job Viewing	The system shall display job description, package, event details, selection process, eligibility and deadline for available drives.
Application	The system shall allow an eligible student to apply to an approved/active drive before its deadline.
Application Status	The system shall allow a student to view the status of submitted applications.
Company Registration	The system shall allow a recruiter to register and maintain company information.
Job Posting	The system shall allow a recruiter to create a placement/job posting.
Job Details	A posting shall support job description, package, event details and deadline.
Eligibility Criteria	A posting shall support eligibility conditions including CGPA, course/department, passing year, backlogs and required skills.
Selection Process	A posting shall support selection stages including GD, project and interview.
Drive Review	The system shall provide TPO with submitted company/job postings for review.
Drive Decision	TPO shall be able to set a drive to Approved, Waiting or Rejected.
Decision Reason	The system shall record a reason/comment when a decision requires justification.

Eligibility Filtering	TPC shall be able to filter students using placement eligibility criteria.
Eligibility Count	The system shall display the count and list of students matching the selected criteria.
Eligibility Engine	The system shall compare student information with the criteria configured for a drive.
Eligibility Result	The system shall provide an eligible/not-eligible result and the relevant criterion information where applicable.
Duplicate Prevention	The system shall prevent duplicate applications for the same student and placement drive.
Deadline Validation	The system shall reject applications submitted after the configured deadline.
Resume Screening	The system shall identify predefined skills/keywords from a submitted resume.
Resume Score	The system shall calculate a basic screening score using predefined matching rules.
Candidate Search	Authorized users shall be able to search/filter candidates using available placement attributes.
Deadline Management	The system shall maintain placement and application deadlines.
Notifications	The system shall provide notifications/messages for important workflow events supported by the implementation.
Director/Dean Information	The system shall provide Director and Dean with authorized placement information.
Message Passing	The system shall support authorized message passing/forwarding according to the finalized workflow.
Feedback	The system shall allow authorized placement participants to submit feedback.
College Reports	The system shall generate college-level placement reports.
Company Reports	The system shall generate company-wise recruitment reports.
Analytics	The system shall provide placement analytics based on stored data.
Audit Logging	The system shall log significant activities including authentication, profile updates, postings, decisions, applications and report generation.
TCP Communication	The system shall exchange client-server requests and responses using TCP sockets.
Concurrent Clients	The server shall support multiple authenticated clients without corrupting shared data.
Persistence	The system shall persist placement data in MySQL through JDBC.

Integrity	The system shall enforce appropriate relational integrity between related records.
Validation and Errors	The system shall validate inputs and return meaningful errors for invalid or unauthorized requests.

6. Data Requirements

Entity	Core Data Elements	Purpose
Student	PRN, Name, Course, Department, CGPA, 10th, 12th, Passing Year, Skills	Eligibility and student management
Company	Company ID, Name, Recruiter/HR details, Industry	Company management
Placement Drive / Job	Drive ID, Company, Title, Description, Package, Event Details, Deadline, Status	Recruitment opportunity
Eligibility Criteria	Drive ID, Minimum CGPA, Course/Department, Backlogs, Passing Year, Skills	Eligibility evaluation
Resume	Resume ID, Student ID, File/Path, Extracted Skills, Score	Resume management/screening
Application	Application ID, Student, Drive, Date, Status	Application tracking
Selection Process	Drive ID, GD, Project, Interview, Other stages	Recruitment process
Feedback	Feedback ID, User/Drive reference, Text, Date	Feedback management
Message/Notification	Sender, Receiver, Message, Timestamp, Status	Communication
Report	Report ID, Type, Filters, Generated By, Date	Reporting
Audit Log	User, Role, Action, Timestamp, Outcome	Traceability

6.1 Data Integrity Rules

- Each major entity shall have a unique identifier.
- Student-to-application and drive-to-application relationships shall be maintained through valid references.
- Duplicate applications for the same student and drive shall be prevented.
- Required fields shall be validated before persistence.
- Deletion/update operations shall preserve referential integrity.

7. External Interface Requirements

7.1 User Interface Requirements

- Login interface with role-aware access.
- Student dashboard for profile, resume, jobs, eligibility and applications.
- Recruiter dashboard for company, job posting, criteria and candidate management.
- TPO dashboard for drive review, decision status, deadlines and reports.
- TPC dashboard for eligibility filtering and eligible counts.
- Director/Dean dashboard for authorized placement information and messaging.
- Administrative interface for system/user management.

7.2 TCP Interface

The communication layer should use a consistent request/response structure. The exact serialization format is an implementation decision unless separately specified.

Logical Field	Description
Operation Type	Identifies requested operation, e.g., LOGIN, DRIVE_CREATE, ELIGIBILITY_CHECK, APPLY or REPORT.
Request ID	Unique identifier for matching a request and response.
User/Session	Authenticated identity/session context.
Role	Role associated with the authenticated session.
Payload	Operation-specific data.
Response Status	Success/failure/result status.
Response Data	Requested result or error information.

7.3 Database Interface

The application shall use JDBC to perform validated database operations against MySQL. SQL operations shall use parameterized statements.

7.4 Resume Interface

The student interface shall support resume upload/update. The exact file formats, maximum size and storage mechanism are implementation details to be finalized before deployment.

8. Non-Functional Requirements

Quality Attribute	Requirement
Performance	Normal user requests should target an average response time below 3 seconds under the defined test environment.
Concurrency	The server shall support multiple authenticated clients concurrently.
Reliability	A client-side failure shall not terminate unrelated active client sessions.
Authentication Security	Protected functions shall require authentication.
Authorization Security	Users shall be restricted to authorized role functions and data.
Credential Protection	Passwords shall not be stored in plain text.
Data Integrity	Transactions and relational constraints shall maintain consistent data.
Usability	The interface shall provide clear navigation, validation and error messages.
Maintainability	The implementation shall use modular, object-oriented components.
Scalability	The system should support growth in students, companies, drives, resumes and applications without fundamental redesign.
Portability	The Java application should run in the supported target operating environments.
Network Robustness	Malformed requests, disconnects and communication errors shall be handled safely.
Auditability	Critical actions shall be recorded with user, time, action and outcome.
Privacy	Student and resume information shall be accessible only to authorized users.
Extensibility	The architecture should permit future modules without major changes to the core workflow.

9. Business Rules

- BR-01: Only authenticated users may access protected placement functions.
- BR-02: A student may apply only when the student satisfies the drive's configured eligibility rules.
- BR-03: Applications submitted after the configured deadline shall not be accepted.

- BR-04: A student shall not have more than one active application for the same drive unless an authorized correction process is defined.
- BR-05: A placement drive requiring TPO approval shall not become available to students until approved.
- BR-06: Rejected drives shall not be available for student applications.
- BR-07: Eligibility shall be determined using the criteria configured for the relevant drive.
- BR-08: Recruiters shall access candidate information associated with their authorized company/drive scope.
- BR-09: Reports shall use authorized and stored placement data.
- BR-10: Director and Dean permissions beyond reading and message passing remain subject to team confirmation.

10. Use Case Specifications

Use Case	Primary Actor	Input	Outcome	Precondition
Login	All Users	User credentials	Authenticated role-based session or error.	User exists and account is active.
Maintain Student Profile	Student	Student data	Profile stored/updated.	Student authenticated.
Upload/Update Resume	Student	Resume file	Resume stored/updated.	Student authenticated.
Register Company	Recruiter	Company details	Company registered.	Recruiter authenticated.
Create Job Posting	Recruiter	Job, package, criteria, deadline	Drive created/submitted.	Company registered.
Review Placement Drive	TPO	Drive details and decision	Approved/Waiting/Rejected	Drive submitted for review.
Filter Eligible Students	TPC/TPO/Authorized Recruiter	Eligibility filters	Eligible list/count.	Drive/criteria available.
View Job	Student	Drive selection	Job details displayed.	Drive is visible to student.
Apply for Drive	Student	Drive ID	Application created.	Student eligible and before deadline.
Screen Resume	Authorized Recruiter/System	Resume and criteria	Matched skills/score.	Resume available.
Update Selection Status	Recruiter/TPO as permitted	Candidate/application	Updated status.	Authorized user.
Generate Report	Authorized User	Report type/filters	Report generated.	User has permission.
Submit Feedback	Authorized Participant	Feedback	Feedback stored.	User authenticated.

Send/Forward Message	Authorized User	Receiver/message	Message recorded/delivered.	Permission granted.
----------------------	-----------------	------------------	-----------------------------	---------------------

11. Reports, Analytics and Feedback

11.1 College Reports

- Registered student count
- Eligible students by company/drive
- Applications by company and department
- Shortlisted/selected/rejected/waiting students
- Department-wise placement statistics
- Placement-drive summary

11.2 Company Reports

- Applicant count
- Eligible candidates
- Resume-screening summary
- Shortlisted candidates
- Application/selection status

11.3 Analytics

Analytics shall summarize placement activity using stored data. Exact KPI formulas and chart types are to be finalized by the project team.

11.4 Feedback

The system shall store relevant placement feedback with appropriate access control.

12. Security and Privacy Requirements

- Passwords shall be securely hashed and never stored as plain text.
- Role-based authorization shall be checked on the server, not only in the client UI.
- Database operations shall use parameterized JDBC statements.
- Inputs shall be validated before processing.
- Resume and student information shall be protected from unauthorized access.
- TCP requests shall be validated for session, operation and payload.
- Internal SQL statements, credentials and stack traces shall not be exposed to users.
- Important actions shall be logged for auditability.

13. System Constraints and Assumptions

13.1 Constraints

- TCP sockets are required for client-server communication.
- MySQL is the relational database.
- Java/JDBC is used for application/database connectivity.
- The portal is designed for a multi-client client-server environment.

13.2 Assumptions

- Users have valid accounts created through the authorized registration/administration process.
- Student academic information supplied to the portal is assumed to be valid unless validation against another system is added later.
- Recruiters provide accurate job and eligibility criteria.
- The server and database are available during normal placement operations.

13.3 Out of Scope for Baseline

- AI/ML-based advanced resume ranking.
- External job-portal integration.
- University ERP integration.
- Virtual interview platforms.
- Mobile application.
- Third-party single sign-on.
- Predictive recruitment analytics.

14. Acceptance Criteria

- Students can authenticate, maintain academic information, upload/update resumes, view jobs and apply.
- Recruiters can register companies, create postings and define eligibility criteria.
- TPO can review drives and assign Approved, Waiting or Rejected status.
- TPC can filter and obtain eligible-student list/count.
- Eligibility results match the configured business rules for test cases.
- Late and duplicate applications are rejected.
- Resume screening produces repeatable results for defined test cases.
- Reports display correct stored data.
- Multiple TCP clients can communicate with the server without cross-user data corruption.
- Role-based access prevents unauthorized operations.
- Database relationships remain consistent after CRUD operations.
- Critical workflow actions are logged.
- Functional, integration and system tests are documented.

15. Conclusion

This SRS establishes the requirements baseline for the Placement Eligibility and Resume Screening Portal. The document should be maintained under version control and updated whenever the approved workflow, user roles, database design, TCP protocol or functional scope changes. The final implementation and test cases should be traceable to the requirement IDs defined in this document.